

Bedrock ESG — Roofing Industry Vertical Specification

DRAFT v0.1 | May 2026

Classification: Pre-Build Specification — Phase 2 Deliverable

1. Executive Summary

This specification defines the integration of the **Roofing Industry Vertical** into the Bedrock ESG platform. The objective is to onboard commercial roofing installations (TPO, EPDM, Metal, and High-Albedo Coatings) as verifiable carbon-offset assets through the Aethexer Protocol.

The core innovation is the **Thermal Load Shedding (TLS)** calculation — a methodology that quantifies the kWh reduction in HVAC demand achieved by high-performance cool roofing systems, converts that reduction into verifiable CO₂e savings, and anchors the proof to Solana via the existing Sovereign Notary.

Note: This vertical does NOT introduce a new token. Verified roofing assets flow through the existing EXITZ tokenization pipeline and the 70/20/10 settlement split.

2. Regulatory & Standards Foundation

All calculations and thresholds in this specification are grounded in published standards. Bedrock ESG does not invent metrics — it automates the verification of metrics that already have institutional backing.

2.1 Applicable Standards

Standard	Authority	Relevance
ASTM C1549-16R22	ASTM International	Solar reflectance measurement (portable reflectometer, 4 wavelengths)
ASTM E1980	ASTM International	Solar Reflectance Index (SRI) calculation
ASHRAE 90.1-2022	ASHRAE	Minimum cool roof requirements: SR $\geq$ 0.70, TE $\geq$ 0.75, or SRI $\geq$ 82
ENERGY STAR (Legacy)	EPA	Low-slope: Initial SR $\geq$ 0.65, 3-year aged SR $\geq$ 0.50; TE $\geq$ 0.75
LEED v4.1 SS Credit	USGBC	SRI $\geq$ 78 (low-slope) or $\geq$ 29 (steep-slope) for 75% of roof area
CRRC-1	Cool Roof Rating Council	Product rating database and aged-value methodology
Title 24 Part 6	California Energy Commission	Climate-zone-specific cool roof requirements

2.2 Performance Tiers

Bedrock ESG defines three performance tiers for roofing assets:

Tier	Solar Reflectance (SR)	Thermal Emittance (TE)	SRI	Classification
Tier 1 — Compliant	SR $\geq$ 0.65	TE $\geq$ 0.75	SRI $\geq$ 78	Meets ENERGY STAR / LEED baseline
Tier 2 — High-Performance	SR $\geq$ 0.80	TE $\geq$ 0.85	SRI $\geq$ 100	Exceeds ASHRAE 90.1 by >10%
Tier 3 — Sub-Ambient Radiative	SR $\geq$ 0.95	TE $\geq$ 0.90 (8–13 μ m window)	SRI $\geq$ 120	Lab-verified radiative cooling membranes

Important: Tier 3 represents emerging radiative cooling technology (e.g., photonic membranes, meta-material coatings). Gemini’s prompt specified $SR \geq 0.97$ — this is achievable only with Tier 3 materials and must not be presented as the baseline requirement. The platform must support all three tiers.

3. Thermal Load Shedding (TLS) Calculation Engine

3.1 Core Formula

The TLS calculation determines the annual HVAC energy reduction attributable to a cool roof installation:

$$TLS \text{ (kWh/year)} = A \times CDD \times Uf \times (1 - SR_{\text{new}}/SR_{\text{baseline}}) \times \eta_{\text{HVAC}}$$

Where:

- **A** = Roof area (m²)
- **CDD** = Cooling Degree Days for the installation’s climate zone (ASHRAE data)
- **Uf** = Roof thermal transmittance (W/m²·K) — derived from R-value
- **SR_new** = Solar reflectance of the installed cool roof
- **SR_baseline** = Solar reflectance of the baseline roof (default: 0.10 for dark asphalt shingle, 0.20 for aged membrane)
- **η_HVAC** = HVAC system efficiency factor (COP of cooling system, default: 3.5)

3.2 CO₂e Conversion

$$CO_2e \text{ Saved (kg/year)} = TLS \text{ (kWh/year)} \times \text{Grid Emission Factor (kg CO}_2\text{e/kWh)}$$

Grid emission factors sourced from EPA eGRID by subregion. Default: **0.386 kg CO₂e/kWh** (US national average 2024).

3.3 Negative Watts (\$NW) — Definition

“Negative Watts” is a Bedrock ESG proprietary metric, **not** an industry standard. It must always be presented with this context.

$$\text{\$NW} = TLS \text{ (kWh/year)} / 8,760 \text{ hours}$$

This yields the average continuous wattage of cooling load avoided — the “negative” power draw. A roof that avoids 43,800 kWh/year = 5 \$NW (5 watts of continuous avoided load).

Compliance Note: \$NW is a communication metric for investor/owner comprehension. All regulatory filings and carbon credit calculations must use standard kWh and kg CO₂e units.

4. Data Input Schema

4.1 Material Registration Fields

Fields required for a roofing company to register their materials in the Bedrock ESG asset registry:

Field	Type	Required	Description
manufacturerName	String	✓	Material manufacturer
productName	String	✓	Commercial product name
batchId	String	✓	Manufacturer's batch/lot ID
substrateType	Enum	✓	TPO , EPDM , METAL_STANDING_SEAM , METAL_CORRUGATED , SPRAY_COATING , PHOTONIC_MEMBRANE
coatingType	String		Coating chemistry (e.g., "Acrylic Elastomeric", "Silicone", "PVDF")
coatingThickness-Mils	Float		Dry film thickness in mils
rValue	Float	✓	Thermal resistance (ft ² ·°F·h/BTU)
solarReflectanceInitial	Float	✓	Initial SR per ASTM C1549 (0.00–1.00)
solarReflectanceAged	Float		3-year aged SR per CRRC methodology
thermalEmittance	Float	✓	TE per ASTM C1371 (0.00–1.00)
sriValue	Float		Solar Reflectance Index per ASTM E1980
crrcProductId	String		CRRC Rated Product ID (if applicable)
spectralTestReportUrl	String		Cloud storage path to spectral test PDF

4.2 Installation Registration Fields

Field	Type	Required	Description
installerCompanyName	String	✓	Licensed roofing contractor
installerLicense	String	✓	State contractor license number
installationDate	DateTime	✓	Date of installation completion
roofAreaSqFt	Float	✓	Total roof area covered
roofSlope	Enum	✓	LOW_SLOPE ($\leq 2:12$) or STEEP_SLOPE ($> 2:12$)
buildingType	Enum	✓	COMMERCIAL , INDUSTRIAL , RESIDENTIAL_HIGH_RISE , WAREHOUSE
climateZone	String	✓	ASHRAE climate zone (e.g., "2A", "3B", "4C")
geoLatitude	Float	✓	Installation latitude
geoLongitude	Float	✓	Installation longitude
previousRoofType	String	✓	Baseline roof being replaced/coated
previousRoofSR	Float		Measured SR of previous roof (default applied if omitted)
hvacSystemType	String		HVAC type for η calculation
hvacCOP	Float		Coefficient of Performance (default: 3.5)
walletAddress	String	✓	Solana wallet for settlement

5. Solana Anchor — Genesis Block Protocol

5.1 Hash Construction

Each verified installation triggers a SHA-256 Genesis Hash anchored to Solana Devnet via the existing Sovereign Notary (`lib/solana-notary.ts`).

```
Genesis Hash = SHA-256(
  batchId +
  solarReflectanceInitial +
  thermalEmittance +
  geoLatitude + geoLongitude +
  installationDate +
  roofAreaSqFt +
  installerLicense
)
```

5.2 Memo Payload Structure

Anchored via the Solana Memo Program (`MemoSq4gqABAXKb96qnH8TysNcWxMyWCqXgDLGmfchR`):

```
{
  "protocol": "aethexer-v1",
  "type": "ROOFING_GENESIS",
  "batchId": "TPO-2026-05-BATCH-4412",
  "genesisHash": "a3f8c2...",
  "tier": "TIER_2",
  "sr": 0.82,
  "te": 0.87,
  "sri": 104,
  "areaM2": 1486.5,
  "climateZone": "2A",
  "coords": [33.4484, -112.0740],
  "sentinel": "9tsq8qB4P9uPRSHrjXLaDQJd93nuo4wEEtkR1tAJRAQ3"
}
```

5.3 Monthly Reporting Cycle

After the Genesis Block, monthly TLS calculations are anchored as `ROOFING_TLS_REPORT` memos:

```
{
  "protocol": "aethexer-v1",
  "type": "ROOFING_TLS_REPORT",
  "genesisHash": "a3f8c2...",
  "reportMonth": "2026-06",
  "tlsKwh": 3650,
  "co2eSavedKg": 1408.9,
  "negativeWatts": 5.0,
  "gridEmissionFactor": 0.386,
  "sentinel": "9tsq8qB4P9uPRSHrjXLaDQJd93nuo4wEEtkR1tAJRAQ3"
}
```

6. Settlement Integration

Roofing settlements use the existing 70/20/10 split with no modifications:

Share	Recipient	Roofing Context
70%	Asset Owner	Building owner or roofing company
20%	Bedrock Treasury	Platform backing / institutional reserve
10%	Specialist	Verification agent who confirmed installation

Settlement is triggered when:

1. Material registration is complete with spectral test data
2. Installation registration is complete with geolocation
3. Genesis Hash is anchored to Solana
4. First monthly TLS report is generated
5. Founder approval gate is passed (human-in-the-loop)

7. Forensic Audit Handshake

The platform verifies installation performance through a three-stage handshake:

Stage 1: Document Forensics

- Spectral test report PDF ingested via `/api/v1/forensic-ingest`
- LLM parser extracts SR, TE, SRI values
- Cross-referenced against CRRC database (if CRRC Product ID provided)
- Parser confidence score generated

Stage 2: Geospatial Verification

- Installation coordinates validated against building footprint data
- Climate zone auto-assigned from ASHRAE lookup table
- Cooling Degree Days pulled from NOAA historical data for the location
- Satellite imagery cross-reference (Phase 3 — future)

Stage 3: Solana Anchor

- Genesis Hash constructed and anchored
- Memo transaction signed by Sentinel Node-01
- `solanaSignature` stored in Settlement record
- Submission status transitions: `DRAFT` → `PENDING_FORENSIC_VERIFICATION` → `FORENSIC_VERIFIED` → `[founderApproved]` → `SETTLEMENT_COMPLETE`

8. Implementation Roadmap

Phase 2A — Schema & API (Estimated: 1 session)

- Add `ROOFING` to `ServiceType` enum
- Add `SubstrateType` enum
- Create `RoofingMaterial` and `RoofingInstallation` models
- Create `/api/roofing/calculate` endpoint (TLS engine)
- Create `/api/roofing/register-material` endpoint
- Create `/api/roofing/register-installation` endpoint

Phase 2B — UI Pages (Estimated: 1 session)

- Add ROOFING section to sidebar navigation
- Build Roofing Calculator page (TLS calculation with SSE streaming)
- Build Material Registry page
- Build Installation Tracker page

Phase 2C — Solana Integration (Estimated: 1 session)

- Extend `anchorToDevnet()` to support `ROOFING_GENESIS` and `ROOFING_TLS_REPORT` memo types
- Wire monthly TLS report generation
- Update Global Ledger to display roofing assets

Phase 2D — Certificates & Compliance (Estimated: 1 session)

- Generate “Thermal Load Shedding Certificate” PDF
 - Include: Installation details, TLS calculation, CO₂e saved, Solana signature, QR code linking to on-chain proof
 - Update Project Bibles with roofing vertical documentation
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9. What This Specification Does NOT Include

- **“Negative Watts” as a tradeable token.** \$NW is a metric, not a token. Verified savings flow through EXITZ.
 - **Elimination claims.** The platform does not claim to “eliminate” HVAC systems. It quantifies reduction in thermal load.
 - **Tier 3 as baseline.** Sub-ambient radiative cooling ($SR \geq 0.95$) is supported but not required. Most commercial installations will be Tier 1 or Tier 2.
 - **Real-time monitoring.** This specification covers document-based verification. IoT sensor integration (roof temperature monitoring) is a Phase 3 consideration.
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10. References

1. ASTM C1549-16R22 — Standard Test Method for Determination of Solar Reflectance Near Ambient Temperature
2. ASTM E1980 — Standard Practice for Calculating Solar Reflectance Index
3. ASHRAE 90.1-2022 — Energy Standard for Buildings Except Low-Rise Residential Buildings

4. ENERGY STAR Roof Products Program Requirements v3.0 (discontinued June 2022, metrics remain industry standard)
 5. LEED v4.1 — Sustainable Sites Credit: Heat Island Reduction
 6. CRRC-1 — Cool Roof Rating Council Product Rating Program
 7. EPA eGRID — Emissions & Generation Resource Integrated Database
 8. PNNL-24904 — Radiative Cooling Roof System Energy Savings Analysis
 9. NOAA — Historical Cooling Degree Day Data by Climate Zone
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